Quiz 1

Name: _____ **Grade:** _____

Instructions. For each multiple-choice question, one or more answers may be correct. Select every correct answer. Question 16 must be answered in the drawing box.

PART 1 — SECURITY FUNDAMENTALS

QUESTION 1

An unauthorized employee reads Alice's medical record without changing or deleting any information. **Which security property is primarily affected?**

- ☐ A. Confidentiality.
- ☐ B. Integrity.
- ☐ C. Availability.
- ☐ D. None of these properties.

QUESTION 2

An attacker changes Alice's laboratory result. The information remains accessible, but its value is now incorrect. **Which security property is primarily affected?**

- ☐ A. Confidentiality.
- ☐ B. Integrity.
- ☐ C. Availability.
- ☐ D. None of these properties.

QUESTION 3

A denial-of-service attack prevents Alice and all other patients from accessing the medical portal. **Which security property is primarily affected?**

- ☐ A. Confidentiality.
- ☐ B. Integrity.
- ☐ C. Availability.
- ☐ D. None of these properties.

PART 2 — COOKIES, ANALYTICS AND ADVERTISING

QUESTION 4

Alice visits `https://shop.example.test` and receives:

```
Set-Cookie: aid=A72; Domain=example.test; Path=/;
            Secure; HttpOnly; SameSite=Lax
```

Which requests will include `aid=A72`? Select all correct answers.

- ☐ A. From `https://shop.example.test`, the browser loads `https://cdn.example.test/banner.png`.
- ☐ B. From `http://shop.example.test`, the browser loads `http://cdn.example.test/banner.png`.
- ☐ C. From `https://social.test`, Alice clicks a link that opens `https://shop.example.test/profile`.
- ☐ D. From `https://social.test`, the page embeds `https://shop.example.test/tracker.png`.

QUESTION 5

When Alice visits `https://shop.example.test`, the shop stores a non-`HttpOnly` first-party cookie named `prefs`. The page loads `metrics.test/analytics.js` and an iframe from `tracker.test`. **Which statements are correct? Select all correct answers.**

- ☐ A. The analytics script executes in the shop's context and can read `prefs`.
- ☐ B. Code inside the iframe can read `prefs` because the iframe is displayed inside the shop's page.
- ☐ C. The request for `analytics.js` automatically sends the shop's `prefs` cookie to `metrics.test`.
- ☐ D. The analytics script can read `prefs` and explicitly send its value to the analytics server.

QUESTION 6

Both `news.alpha.test` and `shop.beta.test` embed a pixel from `tracker.test`. The tracker stores `tid=T9; SameSite=None; Secure`. Assume third-party cookies are allowed. **Which statements are correct? Select all correct answers.**

- ☐ A. Requests sent to `tracker.test` from both publishers can include the same `tid` cookie.
- ☐ B. JavaScript belonging to either publisher can read `tid` using `document.cookie`.
- ☐ C. The same-origin policy prevents the browser from sending `tid` to `tracker.test`.
- ☐ D. The tracker can associate visits to both publishers with the same browser profile.

QUESTION 7

Third-party cookies are blocked. Both `news.test` and `shop.test` load the same analytics script from `metrics.test`. The browser stores `aid=N31` on `news.test` and `aid=S84` on `shop.test`. **Which statements are correct? Select all correct answers.**

- ☐ A. `N31` and `S84` are first-party identifiers for their respective publishers.
- ☐ B. The browser automatically sends both cookies to `metrics.test` in the `Cookie` request header.
- ☐ C. Using the same analytics script means both websites must store the same identifier.
- ☐ D. The analytics script can read the identifier of the current publisher and explicitly send it to `metrics.test`.

QUESTION 8

The browser produces the following exchange:

```
GET https://tracker-a.test/sync     Cookie: aid=A42
302 Location: https://tracker-b.test/sync?aid=A42
GET https://tracker-b.test/sync?aid=A42     Cookie: bid=B71
```

Which conclusions are correct? Select all correct answers.

- ☐ A. Tracker B can associate `A42` with `B71`.
- ☐ B. Tracker B receives `A42` as one of its own cookies.
- ☐ C. Tracker A passes its identifier to Tracker B through the redirected URL.
- ☐ D. The synchronization works only if both trackers have the same origin.

QUESTION 9

During an RTB auction, a publisher sends a bid request through an SSP and ad exchange to several DSPs. It contains the page URL, approximate location and a pseudonymous identifier. DSP-X receives the request but loses the auction. **Which statements are correct? Select all correct answers.**

- ☐ A. The SSP helps the publisher offer its advertising space to potential buyers.
- ☐ B. DSP-X may receive information about the user even though it loses the auction.
- ☐ C. Only the DSP that wins the auction can access the bid request.
- ☐ D. The pseudonymous identifier prevents DSP-X from linking this request with future bid requests.

PART 3 — RESPAWNING AND FINGERPRINTING

QUESTION 10

`C` is the cookie, `L` local storage, `E` the ETag, `F` the fingerprint and `P` the server-side profile.

Test	Action before visiting	C	L	E / F	P
1	Nothing	1234	1234	1234 / 88	3
2	Delete the cookie	1234	1234	1234 / 88	3
3	Delete cookie + local storage	1234	New	1234 / 88	3
4	Clear cookies, storage + cache	5678	5678	5678 / 88	3
5	Use a different device	9012	9012	9012 / 42	9

Which interpretations of Tests 2 and 3 are reasonable? Select all correct answers.

- ☐ A. Test 2 is consistent with restoring the cookie from local storage.
- ☐ B. Test 2 proves that the website used browser fingerprinting.
- ☐ C. Test 3 is consistent with restoring the cookie from the cached ETag.
- ☐ D. Deleting the cookie prevented the server from linking Alice to profile `3`.

QUESTION 11

Using Tests 4 and 5, **which conclusions are reasonable? Select all correct answers.**

- ☐ A. Test 4 proves that fingerprint `88` identifies Alice as a named person.
- ☐ B. Test 4 is consistent with stateless relinking based on fingerprint `88`.
- ☐ C. The new values `5678` prove that cookies, local storage and cache were not cleared.
- ☐ D. Different browsers could share fingerprint `88`, so fingerprint-based linking may produce false positives.

QUESTION 12

Which action is sufficient to prevent the respawning mechanisms observed in both Tests 2 and 3?

- ☐ A. Clear the website's cookies, local storage and relevant cache state together.
- ☐ B. Delete the cookie after every browsing session.
- ☐ C. Clear local storage while keeping the browser cache.
- ☐ D. Configure the cookie with a shorter expiration time.

QUESTION 13

Before any JavaScript executes, the server receives Alice's IP address and an HTTP request containing `User-Agent`, `Accept-Language`, `Accept-Encoding` and browser client hints. **Which statement is correct?**

- ☐ A. The server can observe these values directly from the HTTP request.
- ☐ B. Screen resolution and installed fonts are always included in HTTP request headers.
- ☐ C. Standard HTTP headers cannot contribute to distinguishing one browser from another.
- ☐ D. The server must first store a cookie before it can observe these values.

QUESTION 14

How does Canvas fingerprinting work?

- ☐ A. It stores a random identifier permanently inside the Canvas element.
- ☐ B. It asks the browser to render predefined text and shapes, reads the pixels and hashes the output, which may vary across configurations.
- ☐ C. It uses the Canvas API to retrieve the user's previously viewed images.
- ☐ D. It requests the graphics card's unique hardware serial number through the Canvas API.

QUESTION 15

A dataset contains 100 browser fingerprints: 50 use Windows, 25 use Linux and 25 use macOS. **What is the entropy of the operating-system feature?**

- ☐ A. 0.5 bits.
- ☐ B. 1 bit.
- ☐ C. 1.5 bits.
- ☐ D. 2 bits.

PART 4 — DESIGN CHALLENGE

QUESTION 16

Tracker 1 stores the identifier `1234` in its cookie, while Tracker 2 stores the identifier `5678` in its own cookie. The two trackers want to determine that these identifiers belong to the same browser.

In the box below, draw and explain a detailed cookie-syncing process that allows the two trackers to associate their identifiers.